SIMATS ENGINEERING

CSA17 – ARTIFICIAL INTELLIGENCE LAB

Day 1 – Question 7: 7. Breadth-First Search (BFS)

Aim: To traverse a graph using Breadth-First Search.

Python Program

```
from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])
    while queue:
        node = queue.popleft()
        if node not in visited:
            print(node, end=" ")
            visited.add(node)
            for neighbour in graph[node]:
                if neighbour not in visited:
                    queue.append(neighbour)

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [], 'E': [], 'F': []
}
bfs(graph, 'A')
```

Sample Output

```
A B C D E F
```

Result: The program demonstrates the stated Artificial Intelligence problem.